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Research Article

Title: Loss of species richness with land use intensity is explained by a reduction in niche differences.

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Abstract

Increases in land use intensity (LUI) reduce species richness. However, we have a poor understanding of how underlying coexistence mechanisms are altered by land use and whether diversity loss occurs due to changes in plant-plant interactions (competition and facilitation) or in species intrinsic growth rates. We expect that LUI could reduce stabilizing niche differences and the indirect interactions that promote coexistence (e.g., intransitivity), while increasing competitive inequalities between species. To test the importance of these different processes, we use 8-yr time series from 150 grasslands differing in LUI to evaluate the role of direct and indirect interactions in promoting coexistence between 50 plant species. We show that LUI reduces the number of coexisting species mostly by causing a non-linear reduction in niche differences, rather than by enhancing competitive inequalities. However, surprisingly, niche differences remained important in stabilizing coexistence between those species remaining at high LUI. Indirect interactions were generally less important than direct ones, and played a moderate role in promoting coexistence in smaller assemblages of species at intermediate LUI. Our models could accurately reproduce the decline in diversity seen with LUI, indicating that our time series approach captures the important interactions between species. By analyzing land use effects through recent advances in structural stability applied to community ecology we provide a more mechanistic understanding of its effects. Our results highlight the importance of identifying the niche differences that are lost with increasing LUI, to better predict and manage effects of land use on biodiversity.

Significant statement

Human land use is a major threat to grassland biodiversity. Grasslands with high rates of fertilization, grazing and mowing, contain many fewer plant species. Knowing the underlying causes is necessary for a better management of biodiversity. Here we apply ecological theory to spatiotemporal data on changes in plant abundance in managed grasslands in central Europe. We show that the observed decline in diversity can be explained by how interactions among plant species change with increases in

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land use intensity. In particular, intensive land use removes the stabilizing effect of self-limiting processes that buffer species against extinction as well as limit competitive dominance. Therefore, actions to promote these stabilizing dynamics among interacting species seem key to restore plant diversity.

Introduction

Changes in land use intensity (LUI), for instance increases in fertilization, mowing (e.g., early and frequent cuts), or grazing intensity, are often a major threat to biodiversity (1). Although many studies have shown that plant diversity declines (2–4) and species composition changes (5, 6) as land use is intensified, we have a poor understanding of the mechanisms responsible, which hampers efforts to predict, conserve, and manage land use effects. Several studies have tried to understand mechanisms of community assembly from changes in functional diversity and have shown that functional diversity declines with land use intensification as communities at high land use intensity become dominated by functionally similar species (5, 7, 8). For instance, the reduction of plant canopy height, leaf dry matter content, and seed mass associated with land use intensification has been interpreted as an environmental filtering process (5). However, it can be challenging to infer precise coexistence mechanisms from functional trait distributions (9). An alternative perspective is to take a demographic approach in which changes in population abundance over time are analyzed to estimate the role of species interactions and environmental conditions in determining species coexistence, i.e., the probability that interacting species can persist over time (10–12). In general, land use intensification could alter species intrinsic growth rates, and the strength of competitive and facilitative interactions (13–15) between them, by changing the biotic and abiotic environment. If the altered environmental conditions in intensively managed grasslands result in negative intrinsic growth rates for some species, theory predicts that they will become extinct, with rare species more likely to be lost by chance (5, 10, 16). However, land use intensification can cause further diversity loss by altering species interactions and the balance of intrinsic growth rates between them. Land use intensification can therefore disrupt two main mechanisms of coexistence (17): 1) stabilizing mechanisms, defined as niche differences,

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stabilize the population dynamics of interacting species, by causing negative frequency-dependent growth; and 2) equalizing mechanisms, defined as fitness differences, drive competitive similarity among species when differences in intrinsic growth rates are reduced.

Various aspects of land use intensification may alter niche and fitness differences. Some well documented cases are when fertilization favors fast-growing species, and thereby increases competitive inequalities, by increasing the intrinsic growth of good light competitors at the expense of other species (i.e., LUI could increase fitness differences) (18). These increases in competitive asymmetry make it harder for species to coexist, unless superior competitors suffer self-limitation, which occurs when intraspecific exceeds interspecific competition (i.e., niche differences stabilize competition between species). All processes that cause intraspecific competition to exceed interspecific (e.g., specialized resource uptake, natural enemies, etc.) contribute to these niche differences. Such self-limiting niche differences could also be reduced in intensively-managed grasslands: for instance, fertilizing grasslands could prevent trade-offs in nutrient use from stabilizing coexistence, by reducing the number of resources, such as N, P, and K, that plants compete for (19). These changes would lead to smaller niche differences between plant species, and therefore, less stable coexistence between them (14, 17). On the other hand, diversity loss can lead to an increase in specialist pests and pathogens (20), which might stabilize coexistence between remaining species. Many other changes in coexistence mechanisms are possible and land use intensification could therefore lead to complex changes in multiple mechanisms stabilizing the dynamics of interacting species. However, this range of different mechanisms can be collectively explored by quantifying overall niche differences along gradients of land use intensity.

In multispecies communities, species can also coexist through two types of indirect interactions. The first possibility is intransitive competition, e.g., the rock-paper-scissors game in which species A excludes B, species B excludes C, and species C excludes A, resulting in no universally superior competitor (21). Weaker forms of intransitivity are also possible and could promote coexistence in

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combination with niche differences (22, 23). The second possibility is that a superior competitor differentially harms two inferior competitors and thereby indirectly facilitates one of them (23). For instance, species B and C cannot coexist together alone but if species A harms species B more than C, then it can indirectly facilitate species C and allow the triplet to coexist (24). Land use may reduce the importance of these indirect interactions for coexistence (21, 25): for example, land use intensification can lead to more hierarchical competitive networks (18, 19), in which a few species dominate and exclude the others, because it leads to greater differences in individual size and therefore a greater imbalance in competitive effects (26). In addition, land use could disrupt other types of indirect interactions because it reduces facilitative interactions between species (7, 13).

In sum, land use intensification is expected to reduce coexistence via three main mechanisms: 1) enhancing competitive inequalities between species, 2) destabilizing coexistence by reducing niche differences, and/or 3) reducing the opportunity for indirect interactions to promote species coexistence. No studies have been able to determine the relative importance of these different mechanisms in driving the species loss (i.e., reduction in number of species) frequently observed with land use intensification. A key objective is therefore to better understand how coexistence mechanisms emerging at the pairwise and at the multispecies level vary simultaneously along a LUI gradient and how this contributes to species loss in managed grasslands.

Here, we address this gap by applying a structural stability approach (17, 27) to a unique dataset from the German Biodiversity Exploratories. This includes data on changes in abundance/cover for 50 perennial plant species, across 8 years, in 150 grasslands differing in their degree of land-use intensification (28, 29). In these grasslands, we observe strong declines in species richness with increases in LUI (2, 30) (SI Appendix Fig. S1), similar to the decline in richness seen in a long term fertilization (4). The structural approach allows us to estimate the three types of mechanism that contribute to coexistence, for assemblages of any number of species (see Fig. 1 for a detailed description). Briefly, both niche differences and indirect interactions jointly determine the size of the

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coexistence region (i.e., the feasible domain, Fig. 1), and the larger the region, the more likely species are to coexist. At the same time larger differences in the ability of a species to increase its population (i.e., differences in intrinsic growth rate) make coexistence more difficult (17, 31). We first use the changes in species abundances (i.e., proportion of cover) over time to estimate species intrinsic growth rates, and all pairwise interactions between species, and to assess how LUI changes these growth rates and pairwise interactions. We then couple a population model with the structural stability approach to estimate how land use changes coexistence mechanisms. We do this by calculating the expected structural niche differences, structural fitness differences and indirect interactions for all sets of two and three species, at multiple points along the LUI gradient. To validate our models and to show the importance of these modeling tools for practitioners and conservation biologists, we use them to test whether the predicted number of species that is expected to coexist along the land use intensity gradient is similar to the observed pattern in the decline of number of species with LUI (2, 30) (SI Appendix Fig. S1).

Results and Discussion

Changes in the intensity of grazing, mowing, and fertilization, summarized in our LUI index (28), altered the strength of intra and interspecific competition in opposite ways. Intraspecific competition remained high across the LUI gradient, which should result in the preservation of strong (structural) niche differences and a large coexistence region. However, we also observed complex changes in interspecific competition, which was stronger and more variable at low and particularly at high LUI (SI Appendix, Fig. S2). These combined shifts in intra and interspecific competition resulted in a nonlinear decline in niche differences with LUI, as well as in more variation in niche differences between sets of species (pairs, triplets) at low and high LUI (Fig. 2). In particular, there were more assemblages with very low niche differences at high LUI (e.g. 10% quantile niche differences ~ 0.5 at LUI=0.5 and 3 while niche differences ~ 0.75 at LUI=1.75). The observed change in niche differences was different to that expected by random: a null model in which we shuffled competition coefficients between species showed no variation in the mean trend of niche differences but a marked

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increase in variance across the LUI gradient (SI Appendix Fig. S3). In contrast to our expectations, fitness differences, which arise from differences in intrinsic growth rates between species, did not vary on average across the LUI gradient (Fig. 2). This lack of variation across the LUI gradient was due to the fact that species with higher intrinsic growth rate also showed more sensitivity to interspecific interactions (SI Appendix Fig. S4). We also speculate that this lack of variation could be explained by the fact that fertilization increases fitness differences (18) but at the same time mowing acts as an equalizing mechanism, and grazing reduces fitness differences between fast-growing but palatable versus slow-growing but non-palatable species (32). However, it is challenging to separate these effects with our dataset as mowing and fertilization are strongly correlated with each other (28). Interestingly however, the fitness differences were key drivers of whether assemblages of two and three species could coexist or not, and this effect was consistent along the LUI gradient (Fig. 2). Large differences in competitive ability therefore are the main factor restricting coexistence in general but as they do not change with LUI, changes in competitive ability do not contribute to reduced diversity in intensively managed grasslands. The observed fitness differences also differed from that expected by random. Using our null model, average fitness differences, and the variation between them, increased along the LUI gradient. Because we only randomized interactions coefficients but not intrinsic growth rates, the increase of fitness differences with LUI was due to the fact that we broke the positive relationship between intrinsic growth rate and sensitivity to interspecific interactions (SI Appendix Fig. S4), so species that grow more are no longer those that suffer more at the same time from interspecific interactions. These overall changes in niche and fitness differences were also observed when we accounted for uncertainty in the estimates of intrinsic growth rates and interaction coefficients (SI Appendix Figs. S5 and S6). Taken together, our results show that LUI reduces niche differences more strongly than it increases fitness differences, and this reduction occurs nonlinearly. Such a nonlinear relationship was not predicted by theory, but it implies that coexistence between the 50 most abundant species is actually more likely at low and at intermediate LUI (Fig. 2A and 2C), however, when LUI is increased beyond a certain point, coexistence is strongly reduced.

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It is also worth noting that changes in niche and fitness differences were more pronounced for combinations of three species. In particular, the nonlinear decrease in niche differences with LUI was more pronounced for three (Fig. 2B) than for two species (Fig. 2A). This might suggest that land use intensification does reduce niche differences but this reduction is higher when the assemblage includes a dominant species at high LUI. If dominant species are included this will increase the strength of interspecific effects (this was observed for instance for *Poa trivialis* and *Trifolium pratense*), and therefore, make it more likely to detect a decrease in niche differences. Similarly, we observed an increase in the variance of fitness differences with LUI for sets of three species (Fig. 2D) compared to pairs of species (Fig. 2C). Again, this might imply a divide between a group of particular competitor species and a group of subordinates. These results underline the importance of considering coexistence not only between pairs of species but also between multi species assemblages.

Although niche differences declined with land use intensification, they remained surprisingly high across the LUI gradient, and were often large enough to overcome the observed differences in fitness, at least for pairs of species (Fig. 2). This result suggests that regardless of how grasslands are managed, the species present are not maintained by quasi neutral dynamics, in which intra and interspecific interactions are weak and very similar (33) , and in fact coexistence is maintained by highly structured processes of niche differentiation. We cannot identify the processes that underlie this niche differentiation, but it is likely that in some cases the same processes might maintain coexistence in grasslands differing in land use intensity. For instance, it has been shown that mycorrhizal diversity and abundance do not decline with LUI in these perennial grasslands (34), and mycorrhizae might therefore contribute to coexistence across a range of grasslands (35). In other cases, the mechanisms underlying niche differences might differ between high and low LUI grasslands. For example, foliar pathogen infection and soil fungal pathogen diversity both increase with LUI (34) and specialist fungal pathogens might therefore play a stronger role in driving niche differences in low diversity, high LUI communities. Specialist pathogens could have driven the strong negative frequency dependent population growth seen in *Poa pratensis*, a dominant grass at high LUI. In contrast, at low LUI, we

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might expect trade-offs in nutrient use to play a greater role in driving niche differences (36) . More work is needed to identify the mechanisms underlying the niche differences we find, but our analysis highlights the key importance of niche differences in maintaining coexistence and explaining declines in diversity following land use intensification.

We also found that facilitative (i.e., positive) interactions were common at all land use intensity levels (SI Appendix, Fig. S2). In some cases, they even allowed species to persist in grasslands where they would otherwise have been excluded because the prevailing land use acted as an environmental filter (i.e, intrinsic growth rates below 0) in agreement with suggestions from studies on functional trait distributions (5). For example, the intrinsic growth rate of *Holcus lanatus* became negative from intermediate to high LUI (range -0.17 to -0.35) but the species persisted and our results suggest this could be due to facilitation from several species, including *Lolium perenne* (grass), *Trifolium repens* (legume), and *Plantago lanceolata* (forb) (see Supplementary data). These results showing facilitative interactions in managed grasslands, suggest that facilitation is not restricted to harsh and stressful environments, such as drylands or alpine systems (37–40) and that it may be widespread in many situations. Both niche differentiation and interspecific facilitation have recently been identified as the main drivers of species coexistence in natural ecosystems globally (14, 41) and our results also show that they are the main drivers of species coexistence in heavily human-modified ecosystems. Detailed research is needed to better understand how facilitation operates in intensively managed systems, but our results suggest it could be an important process.

The structural niche differences describe the net effect of both direct and indirect interactions on coexistence. The reduction in niche differences with LUI that we observed for sets of three species (compared to pairs) could arise if LUI more strongly modified indirect interactions than direct pairwise ones. To evaluate the relative importance of indirect interactions in creating (or reducing) novel opportunities for species coexistence, we computed a pair of related metrics for sets of three species. These metrics are community-pair differential and community-pair overlap (see Fig. 1).

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Briefly, community-pair differential is a metric ranging between 1 and -1 that quantifies whether the coexistence region increases or decreases with indirect interactions, i.e., whether indirect interactions promote or reduce coexistence. Community-pair overlap is a metric that quantifies the proportion of the coexistence region for a triplet that is similar to its constituent pairs. It ranges between 0 when triplets and pairs are completely different in their coexistence region, i.e., they coexist in different ways, and 1 when regions are exactly the same.

Neither of the metrics quantifying the effect of indirect interactions on coexistence varied along the LUI gradient in sets of three species, indicating no change in the importance of indirect interactions. Community-pair differential values (which quantify whether indirect interactions decrease or increase the coexistence region) were close to zero (Fig. 3A), indicating that indirect interactions do not enhance coexistence. In addition, community-pair overlap values (the proportion of the coexistence region for a triplet that is similar to its constituent pairs) ranged from 0.5 to 0.8 (Fig. 3B), indicating that the majority of three species combinations can coexist because each of the three pairs that compose the triplet can also do so. These results contrast with prior work done in this system, which suggested a major role for indirect interactions, and particularly intransitivity (21), in promoting species richness. One potential explanation for these contrasting results is that different indirect interactions (e.g., intransitivity vs. indirect facilitation) have contrasting effects on coexistence and cancel out to reveal no overall effect of indirect interactions in this analysis. Another possibility could be that the current analysis included subdominant species, while the previous analysis (21) focused only on intransitivity between dominant species. As intransitive competition has been shown to be stronger between dominant species (21) this might explain the low importance of indirect interactions in the current study. Further work is needed to explore a wider range of indirect interactions and to determine how they change with land use intensification.

It is also possible that indirect interactions become more important for coexistence as we consider larger sets of species. This is because the likelihood of finding a configuration of species interactions

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that maintains biodiversity is higher in larger sets of species. We therefore also computed community-pair differential and community-pair overlap metrics for assemblages of 5, 7, and 11 species, but only focused on the 26 most abundant species in this analysis, to save computing power (see methods). We selected these numbers because it has been shown that indirect interactions promote coexistence when there is an odd number of species in the community, under the assumption of a random tournament (22, 42), and 11 species is on average the maximum number of species found per plot (from the set of 26 species analyzed). We found that community pair differential remained low in combinations of 5, 7, and 11 species. This suggests that indirect interactions did not increase or decrease the opportunities for coexistence amongst larger sets of species (Fig. 3A). However, indirect interactions did change the shape of the feasibility domain, particularly for larger assemblages at intermediate LUI (community-pair overlap at LUI = 1.5 was ~ 0.20 and 0.05 for 7 and 11 species respectively, i.e., 80% and 95% of the coexistence region due to indirect interactions do not overlap with the coexistence region promoted by direct interactions) (Fig. 3B). This indicates that indirect interactions occur in multispecies assemblages and change the way in which the species coexist but they do not lead to more or less coexistence, perhaps because different indirect interactions have opposing effects on coexistence. Overall, these results show that direct pairwise interactions are the main mechanism promoting coexistence across assemblages, however, the opportunities for coexistence that indirect interactions create are completely different compared to those created by direct interactions, particularly at intermediate LUI.

We then analyzed whether we could reproduce the observed decline in species richness with LUI, based on our estimates of how species interactions varied. This allows us to validate our estimates of species interactions by using them to predict diversity. As we found that pairwise interactions were most important for coexistence overall, we focused on the coexistence predicted between species pairs (see the slope of the decline of species richness with LUI, SI Appendix, Fig. S1). We computed the cumulative number of species predicted to coexist in any pairwise combination. For instance, if at LUI = 0.5, species A coexists with B, B excludes C, but species C coexists separately with D, we assume that

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all four species can coexist within the grassland. We then repeated this procedure along the LUI gradient (from 0.5 to 3.0). This approach has been recently shown to reproduce the increase of species richness with area in spatially variable environments because it implicitly allows different species pairs to coexist in different patches (43). Despite the simplicity of the model, we were able to reproduce the decline in species richness with LUI (Fig. 4). Note that we only calculated species richness based on the 50 most common species analyzed in our study, which resulted in an initial slight increase in observed species richness with LUI, followed by a strong decline. However, including all species results in an exponential decline in species richness with LUI because many rare species are only present at the lowest LUI (3). It was not possible to include these rare species in our analysis as we lacked data to estimate changes in their abundance. Further experimental work would be extremely valuable in identifying mechanisms of coexistence between rare species. Overall, we found that the cumulative number of species predicted to coexist across the LUI gradient was close to the observed species richness, and different from randomization predictions which did not capture the reduction in species richness with LUI. Instead the random expectation would be for variation in diversity but no net change with LUI. The only deviation was that we predicted slightly fewer coexisting species than observed at intermediate to high LUI values (LUI=2.25-2.50) (Fig. 4). This difference is most likely due to the reduction in intraspecific competition at intermediate LUI (SI Appendix Fig. S2). Although the comparison between predicted and observed diversity suggests that we might have slightly underestimated the strength of niche differences at intermediate to high LUI, our overall ability to describe the decline in species richness with LUI provides strong evidence that the theoretical approach used here can well describe the observed patterns.

By using time series data from 150 plant communities, we were able to estimate how land use intensification causes declines in plant diversity. Our findings provide strong evidence that a decline in niche differences with LUI explains the loss of diversity and, surprisingly, that changes in competitive ability (intrinsic growth rates) play a relatively minor role. Determining what niche differences are lost as we intensify grassland management is a crucial next step that could enable us to offset the negative

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effects of land use on diversity. We also find that niche differences and interspecific facilitation maintain coexistence even in heavily modified ecosystems. Very little work has been done on how species coexist at high land use intensity, but our results show that, although diversity is reduced, coexistence between the remaining species is highly stable. Uncovering the processes that drive niche differences at high LUI might allow us to better understand and even optimize the ecosystem functioning of intensively managed grasslands. Our coexistence models were able to reproduce the observed diversity declines suggesting our results are robust. The approach of using time series to estimate species interactions and coexistence mechanisms could therefore be widely applied to derive a more mechanistic understanding of global change effects on biodiversity.

Material and methods

Study system

The Biodiversity Exploratories (www.biodiversity-exploratories.de) project (29) is a research initiative that has established 150 permanent grasslands plots of 50 x 50 m in three different regions across Germany: the UNESCO Biosphere Area Schwäbische Alb (south-west), Hainich National Park (central) and the UNESCO Biosphere Reserve Schorfheide-Chorin (north-east). All three regions have a similar climate (a range of 3 °C in mean annual temperature, and 500 to 1000 mm annual precipitation—more details provided in (29)). These 150 plots have been managed as grasslands for at least 20 years prior to the start of the project. Farmers and landowners provide information on the intensity of land management activities, including fertilization, mowing, and grazing. These sites, which range from seminatural to intensively managed grasslands, are either mown or grazed. Grazed plots have cattle, horses, or sheep, with varying numbers of animals and duration of grazing. Grazing intensity was quantified as the number of livestock units ha^{-1} $year^{-1}$. Mown plots are cut one to three times per year. Some grasslands are also fertilized and the intensity is quantified as the amount of organic and inorganic nitrogen added to the grassland (28).

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With this detailed land use information, prior work has quantified a compound index of land use intensity (LUI), which integrates the intensity of fertilization (F), the mowing frequency (M), and the intensity of grazing (G) for each grassland plot (28, 29). For each plot, an individual LUI component (F, M, or G) was standardized relative to its mean across all three regions and all years considered. The compound LUI is the sum of the three standardized components (more details are provided in (3)). The main advantage of this index is to summarize different, correlated aspects of land use into a single metric. We use this index to estimate the effect of human mediated actions on species interaction coefficients and intrinsic growth rates (see next section, *estimation of species interactions*). The minimum LUI of 0.5 could be produced by mowing every 2 y, fertilizing at the rate of 6 kg of N ha^{-1} y^{-1}, or grazing one cow (>2 y old) per hectare for 30 d (or one sheep per hectare for the whole year). An intermediate LUI of 1.5 would equate to around two cuts per year, the addition of 60 kg of N ha^{-1} y^{-1}, or grazing one cow per hectare for most of the year (300 d). A high LUI of 3.0 could be produced by grazing by three cows per hectare for most of the year (300 d) and fertilizing at the rate of 50 kg of N ha^{-1} y^{-1} or by cutting three times and fertilizing with 130 kg of N ha^{-1} y^{-1}. For more details see (3, 29). Within the period of 8 years analyzed, LUI varied in our study between 0.5 to 3.0 across plots (SI Appendix Fig. S1).

Estimation of species interactions

The main aim of our study was to investigate how coexistence mechanisms, at local scales, vary along the LUI gradient. To achieve that aim, we need to estimate species intrinsic growth rates, the matrix of species interactions, including intra and interspecific interactions, and the effects of LUI on both species interactions and intrinsic growth rates. By doing a space for time substitution, we analyzed a time-series of changes in the proportional cover of the 50 most common species (i.e., the most frequently observed species across all plots) between 2008 and 2015. Percentage cover (converted to proportions) was measured for each species and plot in 4m x 4m subplots at peak biomass by Socher and colleagues (44). To obtain estimates of 1) species intrinsic growth rates, 2) pairwise species interaction coefficients, and 3) the effect of LUI on these parameters, we fitted generalized linear

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mixed models (GLMM). Using a similar approach to (45-47), we analysed the log ratio of cover at time *t+1* and cover at time *t* for each focal species (as the dependent variable). The log ratio of cover was modelled as a linear function of LUI, together with the cover of each species in the community (including itself) at time *t*, and we further included the interaction between LUI and cover of all species. The model also included plot nested within region as a random effect and accounted for temporal autocorrelation using an ARMA structure corAR1 (*t+1*). We also tested for autoregressive components with greater time lags (*t+2* and *t+3*) but they resulted in worse statistical fit. Models were fitted using the function *lme* in the R package nlme (48). The general form of the statistical model fitted is as follows:

$$Log\left(\frac{Cover_{i,t+1}}{Cover_{i,t}}\right) \sim r_{i,LUI} + \sum_{j=1}^{n} \alpha_{ij,LUI} * Cover_{j,t}, \text{ random= plot/region + corAR1(year)} \quad (1)$$

In this statistical model, the intercept ($r_{i,LUI}$) was interpreted as the intrinsic growth rate because it estimates the maximum ability of a species to change in cover between two years (t+1/t), at different LUIs, in the absence of any neighbors in the previous year (t), i.e., at intra- and interspecific cover of 0. Because we fitted a separate model for each species *i*, values of the intercept can vary across species and LUI. In turn, the alpha coefficients ($\alpha_{ij,LUI}$) describe how positive (facilitation) or negative (competition) per-capita interactions with conspecific and heterospecific neighbors change with plant cover over time. Again, because our model is fitted for each species *i* independently, it includes the possibility that LUI differently modifies each pairwise interaction coefficient and therefore changes the whole network of species interactions at the community level. Values of species' intrinsic growth rates at different LUI values ($r_{i,LUI}$) are inferred from the regression and lack of data did not allow us to corroborate them empirically. However, we did test the sensitivity of our results to uncertainty in the parameter estimation by calculating coexistence mechanisms (i.e., structural niche and fitness differences) using parameter values at the 2.5% and 97.5% confidence intervals around each parameter (SI Appendix Figs. S5 and S6). We observed that LUI linearly changed intrinsic growth rates for all species. In fact, some intrinsic growth rates were negative at certain LUI values indicating that LUI can

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act as an environmental filter excluding certain species (49). Finally, we did not explicitly model dispersal in our study because there is no information available to assess the influence of propagule pressure on species local abundances (at the plot level). However, previous work in our study system has shown that seed addition does not increase the abundance of the established species, suggesting that seed limitation does not determine population growth rates in these perennial species (50).

Using a structural approach for modeling species coexistence

Following the estimation of species intrinsic growth rates, the matrix of interactions, and the linear effects of LUI on these parameters, we explored how the structural coexistence mechanisms vary along the LUI gradient, using the following discrete time Lotka-Volterra population model (17). We could do so because there is a direct correspondence between the structure of the statistical (equation 1) and the population model (equation 2):

$$log\left(\frac{N_i(t+1)}{N_i(t)}\right) = r_i - \sum_{i=1}^{j} \alpha_{ij} N_j(t) \quad (2)$$

Where N_i and N_j are population sizes of interacting species (measured as proportion of cover), and r_i and α_{ij} are species' intrinsic growth rates and interaction coefficients respectively. Because LUI modified both species' intrinsic growth rates and their pairwise interaction coefficients, we parameterized this model eleven times. Each time we used a different LUI value from 0.5 to 3.0, with a 0.25 LUI increment between parameterizations. This means that we computed structural coexistence mechanisms for each LUI value independently.

The structural approach is a well-established approach in engineering and mathematical sciences to study whether the qualitative behavior of a dynamical system is unaffected by small perturbations. In ecology, it has been recently applied to understand how species within ecological communities persist despite showing variable population dynamics (51–53) but it has not been extended to understand how global change drivers such as land use intensity alter structural coexistence mechanisms. Using this approach has two main advantages over prior work addressing the

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importance of species interactions for coexistence (54): firstly, it allows us to determine how competition, facilitation, and indirect effects jointly combine to maintain biodiversity and second it allows us to explore these ecological mechanisms in a multispecies context. The structural stability approach computes metrics analogous to the pairwise niche differences, which stabilize coexistence between competitors, and the average fitness differences that drive competitive dominance (55, 56). As in the pairwise case, coexistence is possible when structural niche differences exceed structural fitness differences (17, 27). The disadvantage of the approach is that it can be highly demanding of computer power, particularly when considering large numbers of species. For instance, computing all of the coexistence metrics used here, for all possible combinations of the 50 most common species, would take approximately 220 years using 36 CPUs/cores computing in parallel. Given these time limitations, we decided to use the initial pool of 50 species to compute structural niche and fitness differences for all possible combinations of 2 and 3 species only. We then explored the effect of indirect interactions in larger combinations of species, where indirect interactions might be expected to play a larger role. However, we only calculated the measures of indirect interactions (community-pair differential and community-pair overlap) with the 26 most common species (which saves a significant amount of time because there are 4,835 times fewer potential species combinations for 26 than for 50 species). Within the subset of 26 species, the maximum number of species considered in combinations was 11, corresponding to the maximum number of these most common species observed together in a plot.

We first calculated the structural niche differences (SND, hereafter niche differences) and structural fitness differences (SFD, hereafter fitness differences) for all combinations of species pairs and triplets (Fig. 1) and predicted which species combinations are, or are not, stable and feasible (i.e., all species within a combination can coexist because they have abundances greater than 0). Greater niche differences indicate that species are more likely to coexist, whereas greater fitness differences indicate the opposite (17, 27). We then fitted a null model to test whether observed values and outcomes differ from a random expectation. We designed the null model to keep the range and shape of the original

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distribution of species interactions obtained from analyzing the data, while modifying the location of the interaction coefficients within the matrix of species interactions. In other words, we maintained the overall variation in interaction strengths from positive (facilitation) to negative (competition) interactions, because we know this strongly modulates stability conditions (57, 58), but we randomized the strength of interaction coefficients between given species pairs. To do so, we simply reshuffled 100 times the row and column names of the interaction matrix obtained for each LUI value. In this approach species' intrinsic growth rates were not randomized.

We then performed nonlinear quantile regression using the function *nrlq* in the R package quantreg (59) to test the relationships between niche and fitness differences and LUI, both for empirical observations and the null models, and for all combinations of two and three species. Briefly, quantile regressions allow us to explore statistical relationships across the distribution of the response variable (i.e., across quantiles). We took this approach because, after inspecting the scatter plots of niche and fitness differences and LUI, we expected different relationships for the median than for the extremes of the distribution of niche and fitness differences. We used a polynomial regression of the following form: $y \sim a*LUI^2 + b*LUI + c$ where y was either structural niche differences or fitness differences. Results obtained from shuffling interaction coefficients in the randomizations showed different patterns from the observed results (compare observed results in Fig. 2 versus results from randomization in SI Appendix Fig. S3). In particular, in the randomized data we observed no change in niche differences with LUI but a clear increase of fitness differences as well as much greater variation in fitness differences along the LUI gradient. These changes in niche and fitness differences could not explain the decline in species richness with LUI. Importantly, we did not compare niche and fitness differences between combinations of two and three species because these metrics inherently change with increasing species richness in the community.

With the structural analogues of stabilizing niche and fitness differences, we can evaluate the importance of direct pairwise interactions for species coexistence. However, the structural stability

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approach also allows us to test the importance of indirect interactions for multispecies coexistence. To evaluate whether indirect effects increase or reduce species coexistence compared to the pairwise case, we computed a metric called the community-pair differential (Fig. 1). Following Saavedra and colleagues (17), we computed the difference in the size of the feasibility domain for combinations of 3, 5, 7, and 11 species and the feasibility domain for all pairs of species. The community-pair differential ranges between -1 and 1. A positive value indicates greater opportunities for coexistence in the full community (i.e. larger feasibility domain for all species together than for the pairs of species) whereas negative values indicate the opposite. This metric tells us whether indirect interactions create greater opportunities for coexistence by enlarging the feasibility domain. However, it tells us little about whether species actually can coexist thanks to indirect interactions, or if coexistence is achieved by pairwise interactions only (i.e. if the feasibility domain created by pairwise interaction is already sufficient for all species to coexist). Therefore, to evaluate the degree to which indirect, versus pairwise, mechanisms actually explain the coexistence of combinations of 3, 5, 7, and 11 species, we computed a related metric called community-pair overlap. This metric involves calculating the proportion of the feasibility domain for the entire community that lies inside the feasibility domain for all pairs. A value of community-pair overlap close to zero indicates a stronger importance of indirect interactions for species coexistence, whereas a value close to one indicates the opposite (stronger importance of pairwise interactions for coexistence).

After obtaining empirical estimates of all these mechanisms of species coexistence, arising from pairwise and indirect interactions, we finally tested whether our coexistence predictions based on direct interactions between pairs of species are able to reproduce the observed pattern of species diversity decline along the LUI gradient. To make this comparison possible, we calculated the average and standard deviation of the number of species found in each plot versus the maximum number of species predicted to coexist. These predictions were done based on the number of pairs predicted to coexist, with each pair treated independently. For instance, if species A coexists with B, B excludes C,

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but species C coexists separately with D, we assume that all four species can coexist. All analyses were conducted in R version 3.6.3 (60).

Supplementary code and data

R code is located at https://github.com/oscargodoy/LUI coexistence and will be included into an open repository upon acceptance. Plant cover and LUI data are publicly available at https://www.bexis1.uni-jena.de/PublicData/PublicDataSet.aspx?DatasetId=27386 and at https://bdj.pensoft.net/article/36387/, respectively

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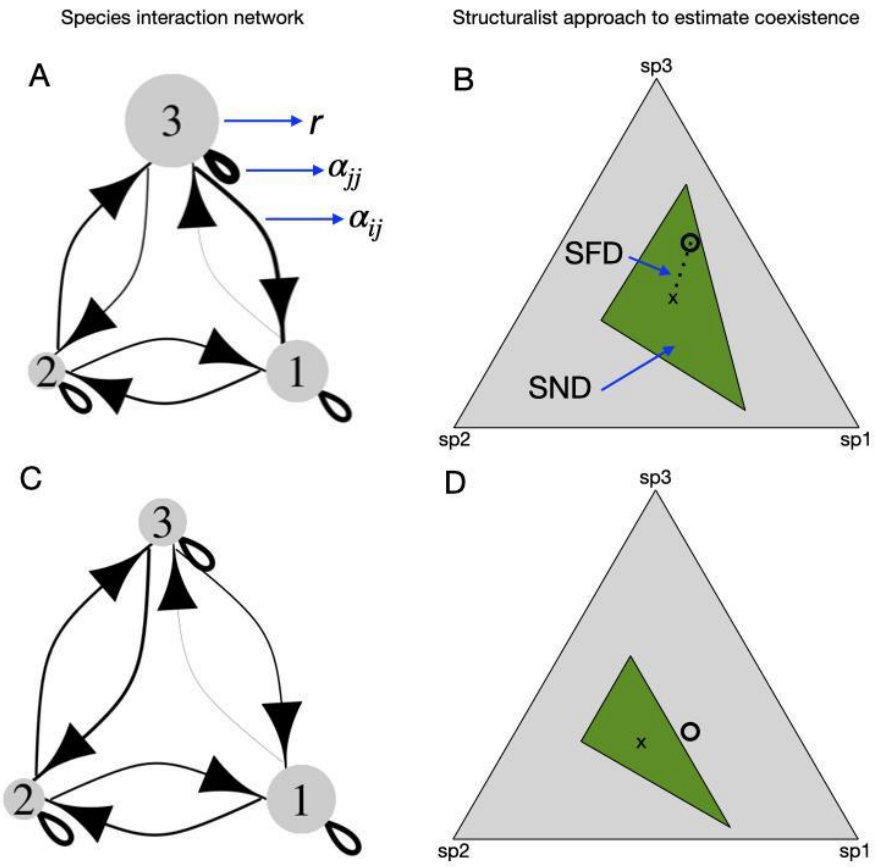
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Figure Legends



Definitions	Formula according to (17)	Range	Ecological interpretation
Structural niche differences (SND, Ω)	$\Omega = \frac{2}{\pi} \arcsin \left(\frac{\alpha_{ii}\alpha_{jj} - \alpha_{ij}\alpha_{ji}}{\sqrt{\alpha_{ii}^2 + \alpha_{ji}^2}\sqrt{\alpha_{jj}^2 + \alpha_{ij}^2}} \right)$	0, ∞	The structure of species interactions determines the size of the feasibility domain (or coexistence region). The larger the domain, the more niche differences there are between the species and therefore, the more likely it is that they can coexist, even if they differ in their intrinsic growth rates.

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Structural fitness differences (SFD, θ)	$\theta = \arccos\left(\frac{r \cdot r_c}{\lvert\lvert r\rvert\rvert \cdot \lvert\lvert r_c\rvert\rvert}\right)$ $r_c = \frac{1}{2}\left(\frac{1}{\sqrt{\alpha_{ii}^2+\alpha_{ji}^2}}\begin{bmatrix}\alpha_{ii}\\ \alpha_{ji}\end{bmatrix} + \frac{1}{\sqrt{\alpha_{ij}^2+\alpha_{jj}^2}}\begin{bmatrix}\alpha_{ij}\\ \alpha_{jj}\end{bmatrix}\right)$	0, ∞	The persistence of all species within a community depends on whether the feasibility domain can accommodate differences in their intrinsic growth rates. Fitness differences are greater if the observed differences in intrinsic growth rates among species (the vector containing each species r) are further from the centroid of the feasibility domain (r_c: this represents the ideal point at which all species have equal intrinsic growth rates). The greater the fitness differences, the less likely that all species can persist together.
Community-pair differential	$\Delta = \Omega - \Omega_{all}$	-1, 1	This metric quantifies the degree to which indirect effects, occurring in multispecies (n>2) assemblages, increase (positive values) or decrease (negative values) the size of the feasibility domain compared to that based on purely pairwise interactions (Ω_{all}), i.e. the feasibility domain based on niche differences alone. Larger domains due to indirect interactions can potentially accommodate larger differences in intrinsic growth rate between species.
Community-pair overlap	$\omega = \frac{D_f}{D_{all}}$	0, 1	This metric quantifies the proportion of the feasibility domain for the entire community (D_f) that lies inside the feasibility domain, under which all species pairs coexist (D_{all}). Therefore, it evaluates whether species mostly coexist due to indirect interactions (values close to zero) or to direct pairwise interactions (values close to one).

Figure 1 Illustration of a three-species interaction network and their persistence according to the structural approach. Panels A and C shows intra-(α_{ii}) and inter-(α_{ij})specific interaction strength between species for two hypothetical cases. In panel A, α_{jj} corresponds to α_{33}, the competitive effect of the species 3 on itself, and α_{ij} corresponds to α_{13}, the competitive effect of species 3 on species 1. Thicker lines indicate stronger competitive effects between species (including intraspecific

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competition). In addition, the size of the circles corresponds to the intrinsic ability of species to grow in the absence of interactions (r), with bigger circles corresponding to higher intrinsic growth rates. Panel B and D shows how the network (matrix) of species interactions and their intrinsic growth rates translate to the drivers of species coexistence using a structuralist approach (structural niche and fitness differences, SND and SFD). The size and the shape of niche differences is the joint result of direct and indirect interactions between species. Higher intraspecific competition compared to interspecific competition results in larger niche differences, and indirect effects can have contrasting effects. We show two contrasted cases. Panel B illustrates a stable and feasible system in which all three species can persist together (i.e., coexist) because niche differences, represented by the green triangle, can accommodate fitness differences, represented by the distance (dashed point line) between the centroid of niche differences (r_c) and the intrinsic growth rates (open circle). Conversely, panel D illustrates the case in which niche differences are not enough to accommodate fitness differences, and therefore, all three species cannot coexist. Formulas below the panel are shown for a two species case to illustrate how intra and interspecific interactions are applied to the calculation of niche and fitness differences, see [7] for a generalization to the n-species case. Figures modified from the shiny app created by William K. Petry (https://ecodynamics.shinyapps.io/StructuralCoexistence/) following Saavedra and colleagues (17) mathematical definitions.

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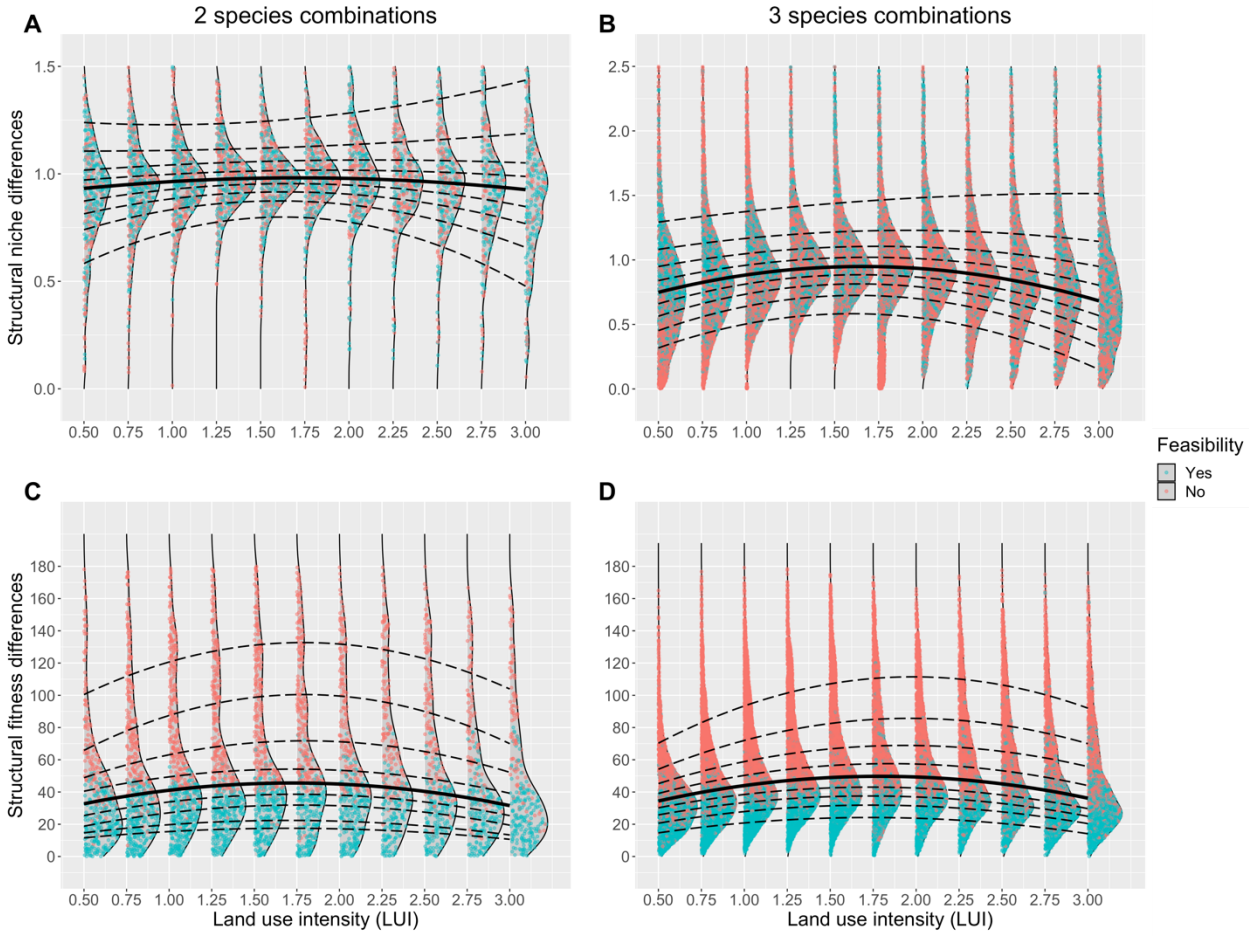


Figure 2 Distribution of structural niche (panels A and B) and structural fitness differences (panels C and D) for two (A and C) and three (B and D) species combinations, across the LUI gradient. Each point corresponds to a species combination and its color denotes whether this combination is predicted to be feasible (blue; all species can coexist) or not (red). The lines across the graph correspond to nonlinear quantile regressions evaluating whether LUI changes structural niche and fitness differences for combinations of two and three species. We performed 9 nonlinear quantile regressions (using a polynomial form $y \sim a*LUI^2 + b*LUI + c$) including the median (thicker solid line) for each panel. Statistical significance is provided in SI Appendix, Table S1.

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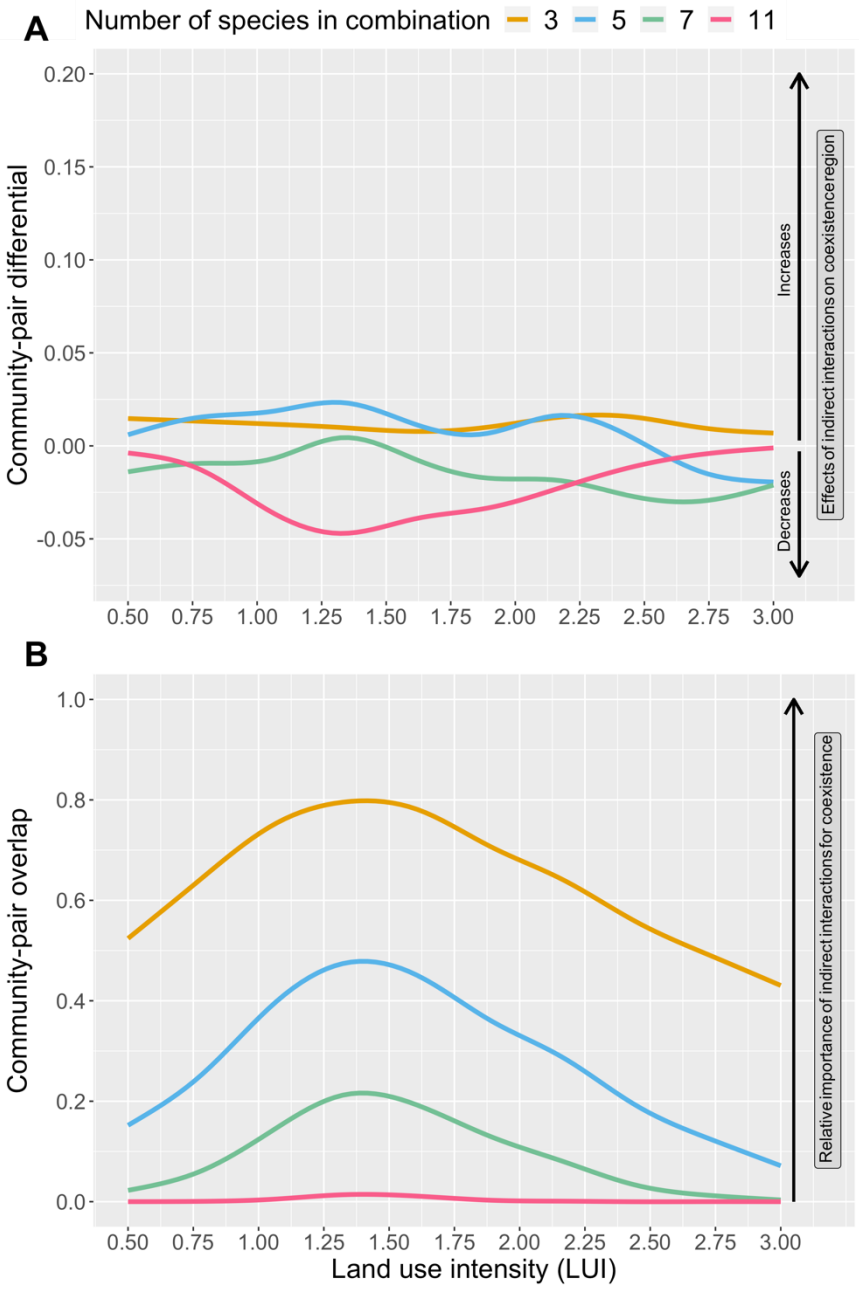


Figure 3 Community- pair differential (i.e., the importance of indirect effects for coexistence, panel A) and community pair overlap (i.e. the importance of pairwise only interactions vs. indirect effects for coexistence, panel B) across the LUI gradient for assemblages of 3, 5, 7, and 11 species. The colors show species assemblages of different sizes. Values correspond to the mean for each number of species. Standard errors are not provided to facilitate visualization.

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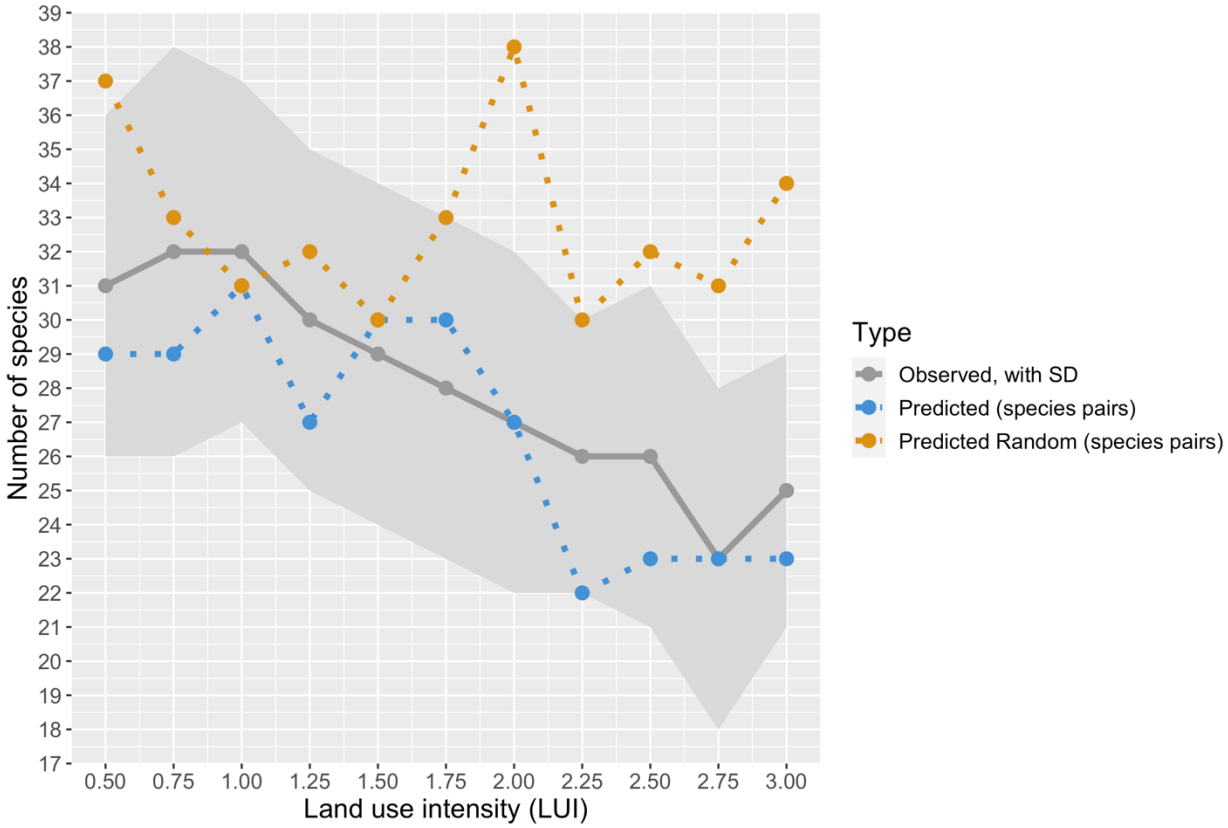
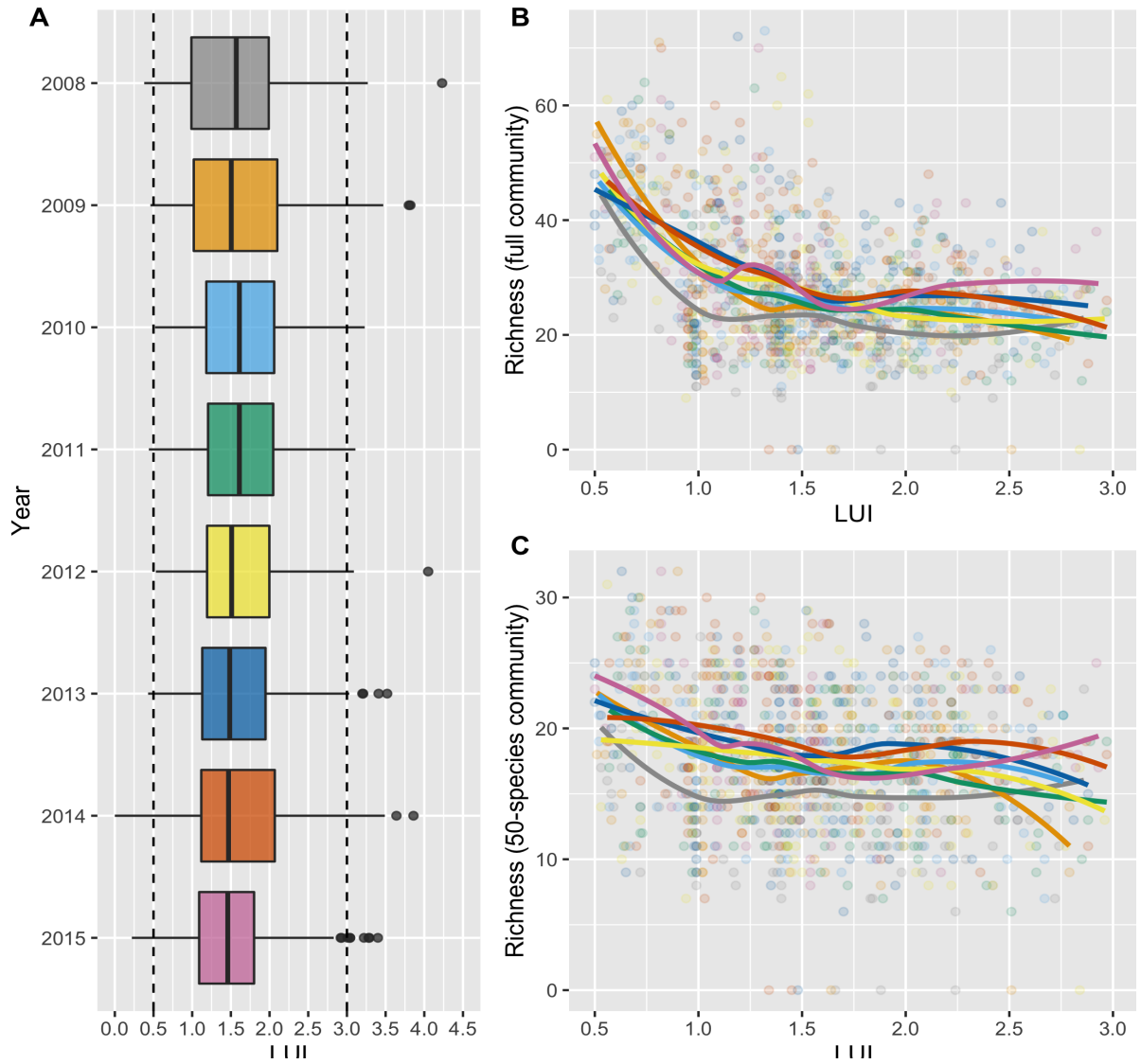


Figure 4 The average number of species (grey line) (+/- standard deviation) observed to co-occur in a plot, the maximum number of species predicted to coexist according to the structural stability approach (in blue), and the maximum number of species predicted to coexist according to null expectations by randomizing species interactions (in orange). Predictions from the models parameterizing species interactions fit well with the observed decline in species richness with LUI. However, randomizing interactions between species leads to much higher coexistence than observed. See methods for details on how the predictions were made.

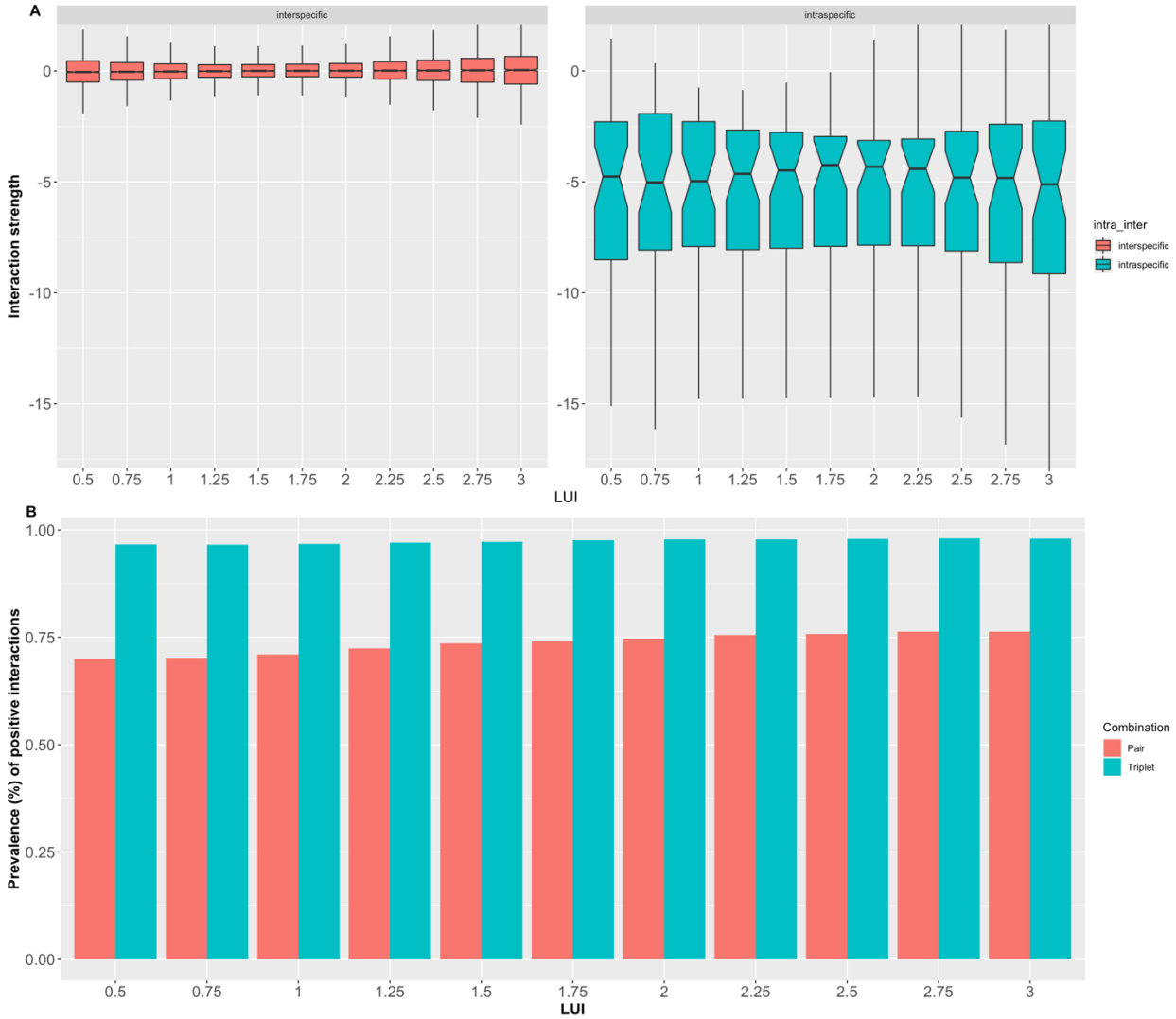
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Supplementary Information Appendix



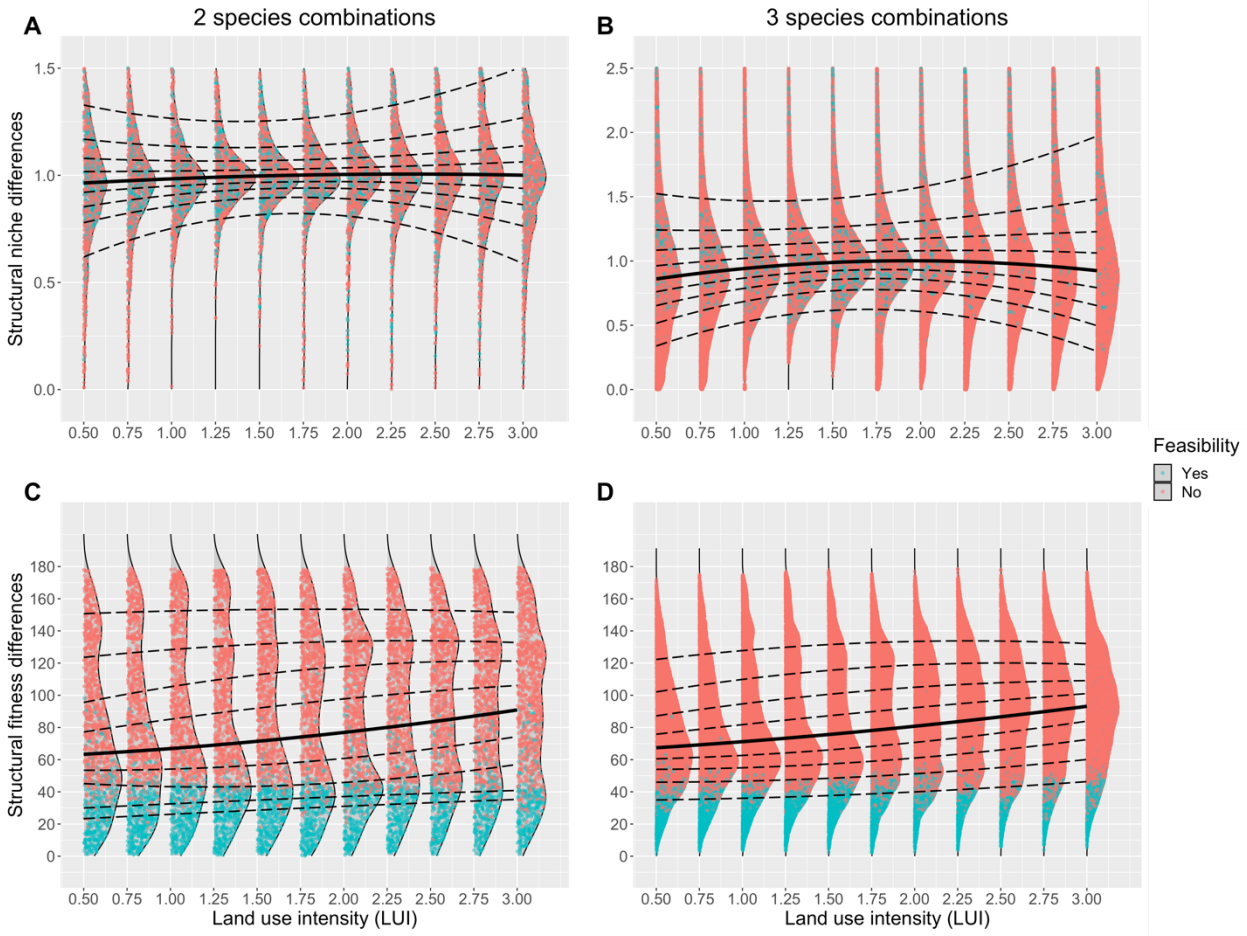
SI Appendix, Figure 1 Range of LUI values for each year considered in the study from 2008 to 2015. According to this information, we decided to take a conservative approach and restrict our analyses to LUI values between 0.5 and 3.0 (Panel A). Variation in richness across the LUI gradient for all species observed in our 150 plots across all years (each color represents a year, Panel B), and variation in richness across the LUI gradient for the 50 most common species selected for the analyses (Panel C).

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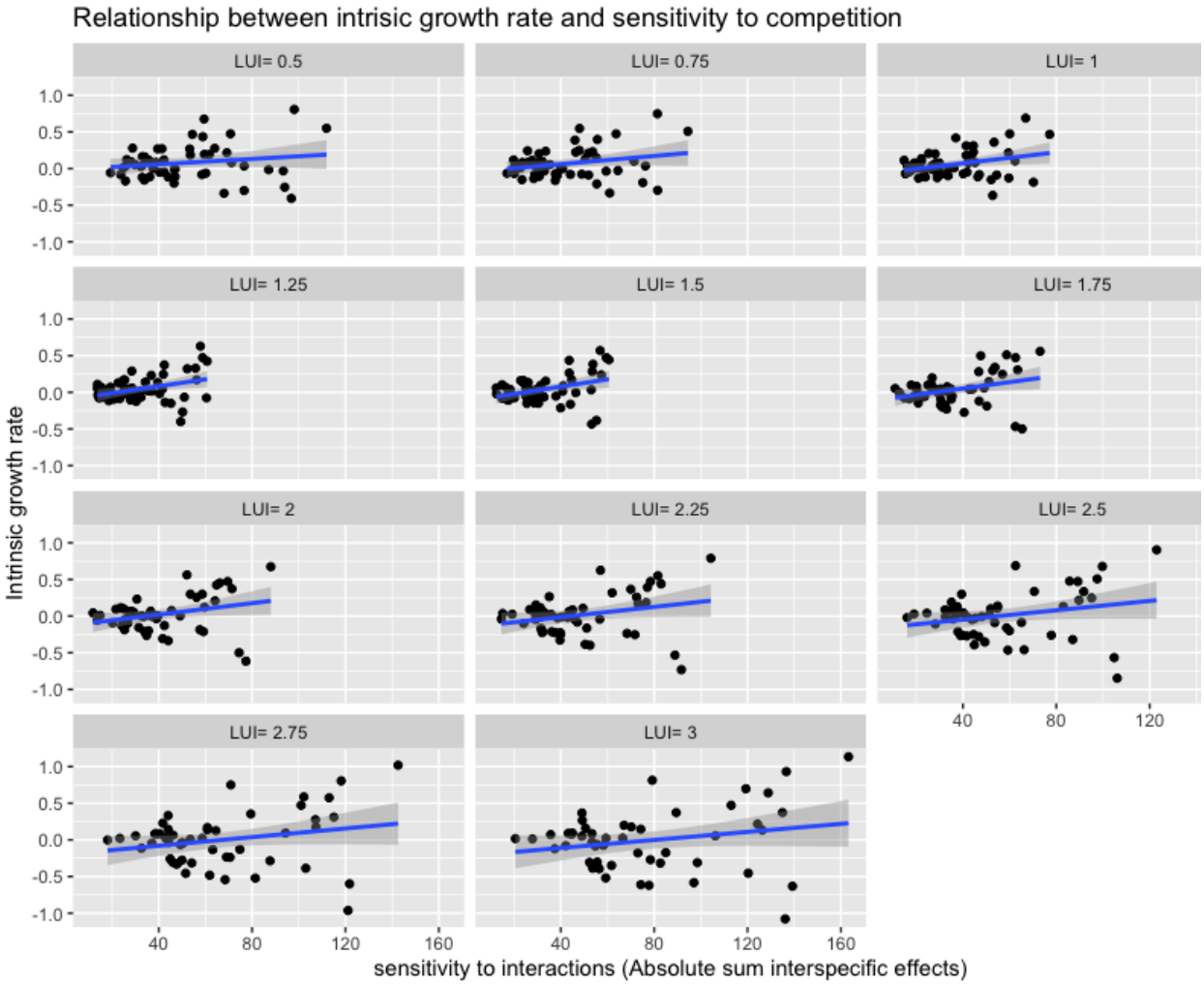
SI Appendix, Figure 2 Land use intensity changes the strength of species interactions. Boxplots show median and range in values for all species at each level of LUI. LUI increases the variance in interspecific competition strength (upper left panel A). LUI also increases the variance of intraspecific competition (intraspecific interactions become more negative as LUI increases) (upper right panel B). Despite these changes, the proportion of combinations of two or three species with at least one positive interaction remained constant across the LUI gradient suggesting that facilitation between species is widespread in our system (bottom panel).

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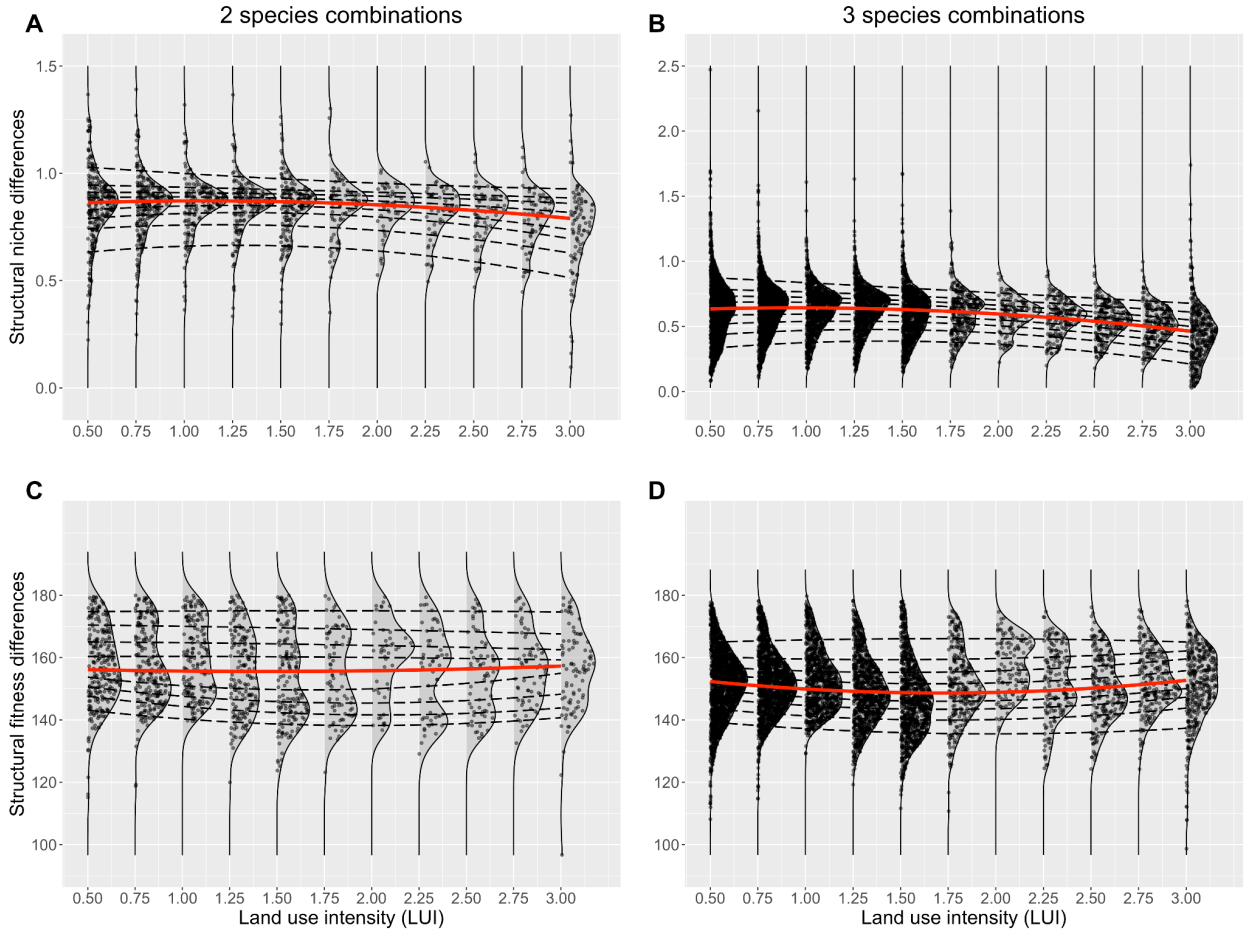
SI Appendix, Figure 3 Mean and standard error values of estimates of quantile regression evaluating whether LUI changes structural niche and fitness differences for combinations of two and three species when values of interaction coefficients were randomized. All quantile regressions were significant. We evaluated 9 quantiles in total from 0.1 to 0.9. (0.5 corresponds to the median). Note that the range of values of the y-axis for both structural niche and fitness differences is similar compared to those presented in the main document Figure 2. Yet there are two main differences. While the structural niche fitness differences remains constant across LUI, structural fitness differences increase under the randomization, and accordingly, the proportion of species combinations predicted to persist is substantially reduced.

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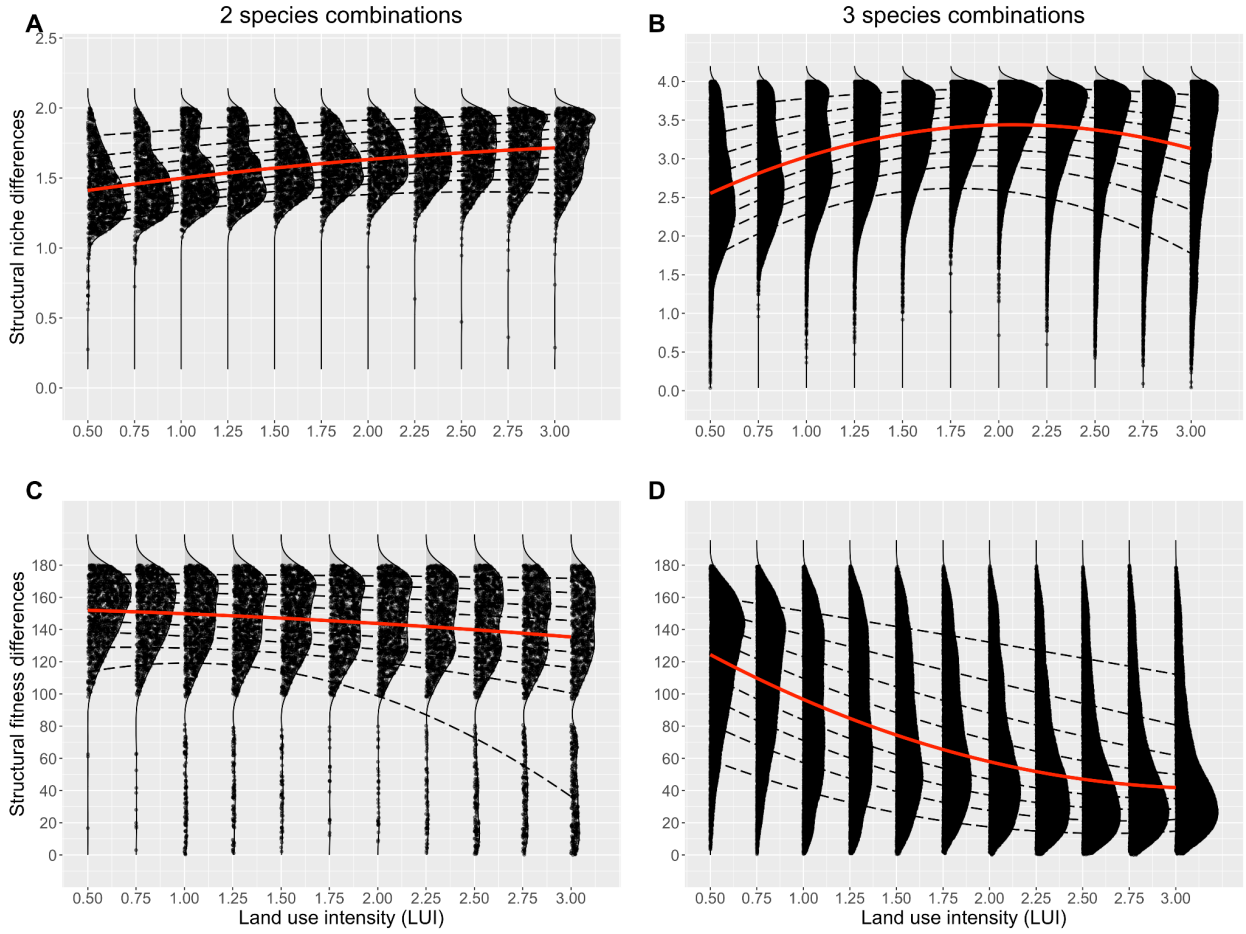
SI Appendix, Figure 4 Relationship between intrinsic growth rate (y axis) and sensitivity to interspecific interactions (x axis). Across the LUI gradient, we observe a positive correlation between species' intrinsic growth rate and their sensitivity to interact measured as the absolute sum by rows of interspecific interaction coefficients. This means that species that grow more are also more susceptible to be affected by competition (or in some cases affected by facilitation) with other species. This trade-off between being a species that grows well but suffers more from competition could explain why we do not find variation in structural fitness differences across the LUI gradient.

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SI Appendix, Figure 5 Distribution of structural niche (panels A and B) and structural fitness differences (panels C and D) taking into consideration the lower estimate of species' intrinsic growth rate and interaction coefficients with a confidence interval of 95%. The distribution of structural niche and fitness differences is shown for two (A and C) and three (B and D) species combinations, across the LUI gradient. Each point corresponds to a species combination. The lines across the graph correspond to nonlinear quantile regressions evaluating whether LUI changes structural niche and fitness differences for combinations of two and three species. We performed 9 nonlinear quantile regressions (using a polynomial form $y \sim a^{*}LUI^{2} + b^{*}LUI + c$) including the median (thicker red solid line) for each panel. Statistical significance is provided in SI Appendix, Table S1.

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SI Appendix, Figure 6 Distribution of structural niche (panels A and B) and structural fitness differences (panels C and D) taking into consideration the upper estimate of species' intrinsic growth rate and interaction coefficients with a confidence interval of 95%. The distribution of structural niche and fitness differences is shown for two (A and C) and three (B and D) species combinations, across the LUI gradient. Each point corresponds to a species combination. The lines across the graph correspond to nonlinear quantile regressions evaluating whether LUI changes structural niche and fitness differences for combinations of two and three species. We performed 9 nonlinear quantile regressions (using a polynomial form $y \sim a^*LUI^2 + b^*LUI + c$) including the median (thicker red solid line) for each panel. Statistical significance is provided in SI Appendix, Table S1.

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SI Appendix, Table S1 Mean ± standard error estimates and associated statistical significance ($p<0.01$, denoted as bold) of the quadratic and the linear coefficients of the quantile regressions.

Quantile regressions associated with Figure 2 Main text

Quantile	2 species, Structural Niche Differences		2 species, Structural Fitness Differences		3 species, Structural Niche Differences		3 species, Structural Fitness Differences	
	Quadratic	Linear	Quadratic	Linear	Quadratic	Linear	Quadratic	Linear
10	**-0.164 ± 0.0160**	**0.530 ± 0.053**	**-1.286 ± 0.500**	**4.912 ± 1.799**	**-0.288 ± 0.006**	**0.924 ± 0.022**	**-3.213 ± 0.196**	**11.974 ± 0.705**
20	**-0.085 ± 0.008**	**0.254 ± 0.030**	**-2.149 ± 0.553**	**8.478 ± 1.913**	**-0.220 ± 0.005**	**0.674 ± 0.018**	**-3.567 ± 0.184**	**12.818 ± 0.643**
30	**-0.054 ± 0.0046**	**0.157 ± 0.014**	**-2.066 ± 0.635**	**8.741 ± 2.298**	**-0.160 ± 0.004**	**0.475 ± 0.016**	**-3.411 ± 0.185**	**12.171 ± 0.648**
40	**-0.033 ± 0.003**	**0.102 ± 0.012**	**-2.332 ± 0.742**	**9.180 ± 2.670**	**-0.121 ± 0.003**	**0.352 ± 0.011**	**3.218 ± 0.149**	**11.309 ± 0.543**
50	**-0.020 ± 0.061**	**0.061 ± 0.011**	**-2.284 ± 0.602**	**8.277 ± 2.133**	**-0.091 ± 0.002**	**0.262 ± 0.009**	**-2.991 ± 0.141**	**10.478 ± 0.490**
60	**-0.010 ± 0.003**	**0.030 ± 0.012**	**-2.064 ± 0.457**	**7.821 ± 1.518**	**-0.065 ± 0.003**	**0.187 ± 0.010**	**-2.466 ± 0.167**	**8.855 ± 0.568**
70	-0.003 ± 0.005	0.005 ± 0.017	**-1.556 ± 0.481**	**5.743 ± 1.687**	**-0.038 ± 0.003**	**0.103 ± 0.012**	**-2.051 ± 0.178**	**7.674 ± 0.591**
80	**0.026 ± 0.009**	**-0.096 ± 0.030**	**-0.623 ± 0.480**	**2.421 ± 1.680**	-0.001 ± 0.004	-0.005 ± 0.015	**-1.280 ± 0.238**	**5.486 ± 0.755**
90	**0.082 ± 0.016**	**-0.272 ± 0.060**	-0.275 ± 0.659	1.452 ± 2.148	**0.062 ± 0.009**	**-0.196 ± 0.030**	0.091 ± 0.307	2.347 ± 1.008

Quantile regressions with confidence interval (lower and upper) associated with Figure S4 and Figure S5 Supplementary information

- Lower confidence interval (Fig. S4)

Quantile	2 species, Structural Niche Differences		2 species, Structural Fitness Differences		3 species, Structural Niche Differences		3 species, Structural Fitness Differences	
	Quadratic	Linear	Quadratic	Linear	Quadratic	Linear	Quadratic	Linear
10	**-0.051 ± 0.015**	0.132 ± 0.055	**2.319 ± 0.637**	**-9.085 ± 2.024**	**-0.069 ± 0.006**	**0.194 ± 0.023**	**1.747 ± 0.358**	**-6.931 ± 1.122**
20	-0.040 ± 0.016	0.095 ± 0.056	**2.509 ± 0.713**	**-10.285 ± 2.389**	**-0.062 ± 0.006**	**0.167 ± 0.021**	**2.424 ± 0.318**	**-8.442 ± 1.051**
30	**-0.042 ± 0.011**	**0.110 ± 0.036**	**2.509 ± 0.846**	**-9.709 ± 2.761**	**-0.053 ± 0.006**	**0.127 ± 0.022**	**2.394 ± 0.332**	**-8.260 ± 1.067**
40	**-0.037 ± 0.007**	**0.093 ± 0.025**	**2.996 ± 1.082**	**-9.986 ± 3.693**	**-0.044 ± 0.005**	**0.089 ± 0.017**	**2.504 ± 0.256**	**-8.647 ± 0.874**

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50	**-0.022 ± 0.007**	0.049 ± 0.024	0.689 ± 1.047	-1.957 ± 3.783	**-0.042 ± 0.004**	**0.080 ± 0.014**	**2.484 ± 0.315**	**-8.545 ± 0.976**
60	**-0.017 ± 0.006**	0.032 ± 0.019	-0.230 ± 0.954	0.377 ± 3.234	**-0.037 ± 0.004**	**0.058 ± 0.014**	**2.347 ± 0.280**	**-7.799 ± 0.932**
70	-0.008 ± 0.006	0.010 ± 0.020	-0.240 ± 0.896	-0.161 ± 3.050	**-0.026 ± 0.005**	0.017 ± 0.016	**1.906 ± 0.304**	**-5.689 ± 1.029**
80	-0.001 ± 0.006	-0.018 ± 0.022	-0.163 ± 1.095	-0.573 ± 3.746	**-0.014 ± 0.004**	-0.021 ± 0.013	**1.126 ± 0.355**	-3.212 ± 1.262
90	0.009 ± 0.011	-0.072 ± 0.038	-0.220 ± 0.745	0.733 ± 2.421	-0.001 ± 0.007	**-0.079 ± 0.024**	-0.713 ± 0.422	2.525 ± 1.439

- Upper confidence interval (Fig. S5)

Quantile	2 species, Structural Niche Differences		2 species, Structural Fitness Differences		3 species, Structural Niche Differences		3 species, Structural Fitness Differences	
	Quadratic	Linear	Quadratic	Linear	Quadratic	Linear	Quadratic	Linear
10	**-0.054 ± 0.004**	**0.280 ± 0.014**	**-20.993 ± 1.233**	**42.534 ± 4.007**	**-0.560 ± 0.005**	**1.988 ± 0.018**	**10.063 ± 0.201**	**-52.904 ± 0.809**
20	**-0.050 ± 0.004**	**0.280 ± 0.014**	**-5.347 ± 1.081**	7.405 ± 3.271	**-0.476 ± 0.004**	**1.814 ± 0.015**	**12.070 ± 0.181**	**-65.881 ± 0.702**
30	**-0.040 ± 0.004**	**0.255 ± 0.014**	-1.617 ± 0.658	-3.293 ± 2.193	**-0.428 ± 0.004**	**1.699 ± 0.015**	**12.730 ± 0.201**	**-72.156 ± 0.764**
40	**-0.031 ± 0.005**	**0.233 ± 0.020**	-1.147 ± 0.607	-3.680 ± 2.035	**-0.393 ± 0.003**	**1.603 ± 0.013**	**12.417 ± 0.203**	**-74.059 ± 0.762**
50	**-0.025 ± 0.005**	**0.210 ± 0.018**	-1.151 ± 0.669	-2.609 ± 2.278	**-0.358 ± 0.003**	**1.487 ± 0.013**	**11.293 ± 0.236**	**-72.558 ± 0.850**
60	**-0.024 ± 0.004**	**0.207 ± 0.015**	-0.934 ± 0.628	-1.632 ± 2.004	**-0.317 ± 0.003**	**1.331 ± 0.013**	**8.475 ± 0.214**	**-63.710 ± 0.755**
70	**-0.016 ± 0.005**	**0.178 ± 0.020**	**-1.413 ± 0.463**	1.207 ± 1.597	**-0.264 ± 0.003**	**1.116 ± 0.013**	**4.388 ± 0.271**	**-48.163 ± 0.902**
80	**-0.017 ± 0.005**	**0.162 ± 0.019**	-1.208 ± 0.471	2.047 ± 1.561	**-0.193 ± 0.003**	**0.817 ± 0.012**	**0.740 ± 0.267**	**-30.970 ± 0.882**
90	**-0.013 ± 0.005**	**0.110 ± 0.020**	-0.290 ± 0.369	-0.029 ± 1.273	**-0.104 ± 0.002**	**0.441 ± 0.008**	-0.343 ± 0.291	**-18.238 ± 0.898**